

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

x	y
1	11
2	19
3	a

The table shows three values of x and their corresponding values of y for the equation $y = 4(2)^x + 3$. In the table, a is a constant. What is the value of a ?

Answer

- A. 67
- B. 35
- C. 32
- D. 27

Correct Answer: B

Rationale

Choice B is correct. It's given that the table shows three values of x and their corresponding values of y for the equation $y = 4(2)^x + 3$. It's also given that when $x = 3$ the corresponding value of y is a , and a is a constant. Substituting 3 for x and a for y in the given equation yields $a = 4(2)^3 + 3$, or $a = 35$. Therefore, the value of a is 35.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.